

Formula Sheet

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$$\hat{p} = \frac{x}{n}$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$t_{pooled} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$t_{Welch} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

$$t_{paired} = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$$

$$\text{where } \bar{d} = \frac{\sum d_i}{n}, \quad s_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n - 1}}$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}(1 - \bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (P_1 - P_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}}$$

$$F = \frac{s_1^2}{s_2^2}$$

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

$$\bar{d} \pm t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}}$$

$$\frac{(n-1)s^2}{\chi_{\alpha/2, v}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\alpha/2, v}^2}$$

$$\frac{s_1^2}{s_2^2} \frac{1}{F_{\alpha/2, df1, df2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} F_{\alpha/2, df2, df1}$$

$$\chi^2 = \sum \frac{(f_0 - f_e)^2}{f_e}$$

$$E_{ij} = \frac{(\text{Row Total}_i \times \text{Col Total}_j)}{\text{Grand Total}}$$

$$df = (R - 1)(C - 1)$$

$$SSC = \sum_{j=1}^C n_j (\bar{x}_j - \bar{\bar{x}})^2$$

$$SSE = \sum_{j=1}^C \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2$$

$$SST = \sum_{j=1}^C \sum_{i=1}^{n_j} (x_{ij} - \bar{\bar{x}})^2$$

$$df_C = C - 1$$

$$df_E = N - C$$

$$df_T = N - 1$$

$$\begin{aligned}
MSC &= \frac{SSC}{df_C}, \quad MSE = \frac{SSE}{df_E} \\
F &= \frac{MSC}{MSE} \\
SSC &= R \sum_{j=1}^C (\bar{x}_{.j} - \bar{\bar{x}})^2 \quad (\text{Columns}) \\
SSR &= C \sum_{i=1}^R (\bar{x}_{i.} - \bar{\bar{x}})^2 \quad (\text{Rows}) \\
SST &= \sum_{i=1}^R \sum_{j=1}^C (x_{ij} - \bar{\bar{x}})^2 \\
SSE &= SST - SSC - SSR \\
df_C &= C - 1, \quad df_R = R - 1 \\
df_E &= (R - 1)(C - 1), \quad df_T = N - 1 \\
MSC &= \frac{SSC}{df_C}, \quad MSR = \frac{SSR}{df_R} \\
MSE &= \frac{SSE}{df_E} \\
F_C &= \frac{MSC}{MSE}, \quad F_R = \frac{MSR}{MSE} \\
CF &= \frac{(T_{..})^2}{n} \\
TSS &= \sum \sum X_{ij}^2 - CF \\
CSS &= \sum \frac{T_{.j}^2}{r} - CF \\
RSS &= \sum \frac{T_{i.}^2}{C} - CF \\
R^2 &= \frac{a \sum Y + b \sum XY - \frac{(\sum Y)^2}{n}}{\sum Y^2 - \frac{(\sum Y)^2}{n}} \\
t &= \frac{a - \alpha}{s_a}
\end{aligned}$$

$$\begin{aligned}
\hat{y} &= a + bx \\
b &= \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}} \\
a &= \bar{y} - b\bar{x} \\
b &= \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2} \\
R^2 &= 1 - \frac{SSE}{SS_{yy}}, \text{ or } R^2 = \frac{SSR}{SS_{yy}} \\
Adj R^2 &= 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1} \\
t &= \frac{b - \beta}{s_b}, \quad s_b = \frac{s_e}{\sqrt{SS_{xx}}} \\
SS_{yy} &= \sum y^2 - \frac{(\sum y)^2}{n} \\
SS_{yy} &= SSR + SSE \\
SSR &= \sum (\hat{y} - \bar{y})^2 \\
SS_{xx} &= \sum x^2 - \frac{(\sum x)^2}{n} \\
s_e &= \sqrt{\frac{SSE}{n - 2}}, \quad SSE = \sum (y - \hat{y})^2 \\
s_a &= S_{yx} \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum (X - \bar{X})^2}} \\
s_b &= \frac{S_{yx}}{\sqrt{\sum (X - \bar{X})^2}} \\
R^2 &= \frac{\sum (\hat{Y} - \bar{Y})^2}{\sum (Y - \bar{Y})^2} \\
r &= \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}} \\
r &= \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{(n \sum x^2 - (\sum x)^2) (n \sum y^2 - (\sum y)^2)}} \\
\text{Moving Avg: } MA_t &= \frac{\sum_{i=0}^{k-1} Y_{t-i}}{k}
\end{aligned}$$